

Circuit & Lab CSE2120 2023 Fall

3 Credit Hours (Elective course for CSE)

(※ Mandatory course for ICE)

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Textbooks:

1. Nilsson and Riedel, Electric Circuits (10th Ed.), Prentice Hall, 2015.

Chapter 1	Circuit Variables	2
Chapter 2	Circuit Elements	24
Chapter 3	Simple Resistive Circuits	56
Chapter 4	Techniques of Circuit Analysis	88
Chapter 5	The Operational Amplifier	144
Chapter 6	Inductance, Capacitance, and Mutual Inductance	174
Chapter 7	Response of First-Order <i>RL</i> and <i>RC</i> Circuits	212
Chapter 8	Natural and Step Responses of <i>RLC</i> Circuits	264
Chapter 9	Sinusoidal Steady-State Analysis	304
Chapter 10	Sinusoidal Steady-State Power Calculations	358
Chapter 11	Balanced Three-Phase Circuits	396
Chapter 12	Introduction to the Laplace Transform	426
Chapter 13	The Laplace Transform in Circuit Analysis	464
Chapter 14	Introduction to Frequency Selective Circuits	520
Chapter 15	Active Filter Circuits	556
Chapter 16	Fourier Series	602
Chapter 17	The Fourier Transform	642
Chapter 18	Two-Port Circuits	676

2. Boylestad and Nashelsky, Electronic Devices and Circuit Theory (11th Ed.), Pearson, 2014.

CHAPTER 1: Semiconductor Diodes	1
CHAPTER 2: Diode Applications	55
CHAPTER 3: Bipolar Junction Transistors	129
CHAPTER 4: DC Biasing–BJTs	160
CHAPTER 5: BJT AC Analysis	253
CHAPTER 6: Field-Effect Transistors	378
CHAPTER 7: FET Biasing	422
CHAPTER 8: FET Amplifiers	481
CHAPTER 9: BJT and JFET Frequency Response	545
CHAPTER 10: Operational Amplifiers	607
CHAPTER 11: Op-Amp Applications	653
CHAPTER 12: Power Amplifiers	683
CHAPTER 13: Linear-Digital ICs	722
CHAPTER 14: Feedback and Oscillator Circuits	751
CHAPTER 15: Power Supplies (Voltage Regulators)	783
CHAPTER 16: Other Two-Terminal Devices	811
CHAPTER 17: <i>pnpn</i> and Other Devices	841

Reference Books

1. Hayt, Kemmerly and Durbin, Engineering Circuit Analysis (9th Ed.), McGraw-Hill, 2019.
2. Nahvi and Edminister, Schaum's Outlines of Electric Circuits, McGraw-Hill, 2018.
3. Sedra and Smith, Microelectronic Circuits (7th Ed.), Oxford Univ, 2015.

REFERENCES in Internet

1. MIT OpenCourseWare (OCW)

https://ocw.mit.edu/search/?d=Electrical%20Engineering%20and%20Computer%20Science&s=department_course_numbers.sort_coursenum

* <https://ocw.mit.edu/>

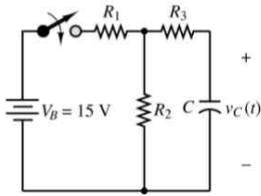
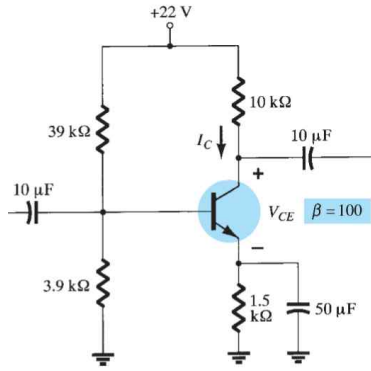
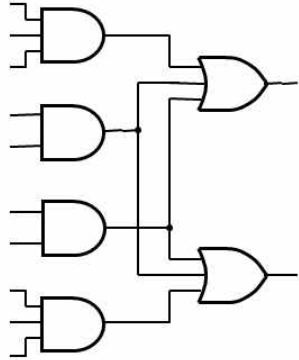
2. Edison Tech Center

<https://edisontechcenter.org/HallofFame.html>

What will we study in the Circuit and Lab course?

Analog Hardwares	1. Electric Circuits
	2. Electronic Circuits

Differences

Electric Circuits (R, L, C)	Electronic Circuits (R, L, C, Diodes, BJTs, FETs)	Digital Circuits (Gates)
		
differential equation Laplace transform complex number	Laplace transform	Boolean Algebra (0/1 : very easy)

Grading

Midterm Exam	40 %
Final Exam	45 %
Attendance	5 %
Homeworks	10 %

A: 20 ± 5 % B: 30 ± 5 % C: 30 ± 5 % D&F: 20 ± 5 %

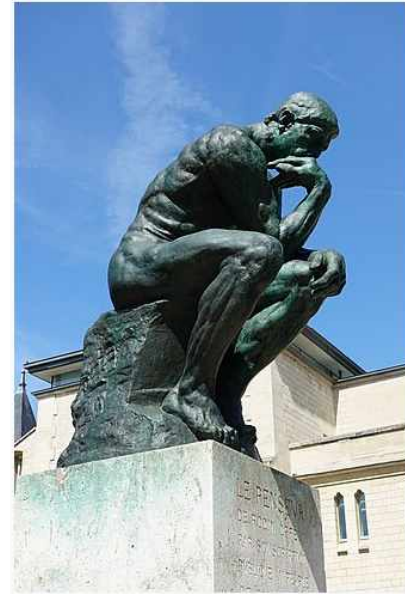
Homework Policies

- All homeworks must be done by yourself.
(Do not copy, please. If copy, you will get ZERO.)
- Upload homeworks on the eclass within the first due dates.
(within the second due dates : 80%)
- TA will grade homeworks.

How to Study

- If you have any question :
 1. Read textbooks and reference books thoroughly FIRST.
 2. THINK, THINK, THINK.
 3. Try to get solutions by yourself.
 4. THINK, THINK, THINK.
 5. Get some information from internet.
 6. THINK, THINK, THINK.
 7. Discuss with your friends.
 8. THINK, THINK, THINK.
 9. Try to get solutions by yourself, again and again.
 10. THINK, THINK, THINK.
 11. Finally, visit the professor's office.

The Thinker - Rodin



※ Make a STUDY GROUP

BEFORE THINKING, DO NOT ASK

MUST change the studying methods

(You are not a high school student, but a university student.)

IN UNIVERSITY, YOU MUST STUDY HOW TO THINK LOGICALLY.

(Differences between Engineers and Technicians)

※ English

British English, American English, Uzbek English,
Russian English, Korean English, Japanese English,
Chinese English,

Scholarships

<https://www.afterschoolafrica.com/>

Korean Scholarships

<https://www.afterschoolafrica.com/5720/korean-government-scholarship-bachelors-masters-phd-developing-countries/>

<https://en.kf.or.kr/>

<https://www.studyinkorea.go.kr/en/main.do>

※ The best way for Master and Ph.D. degrees in Korea

Contact professors in Korean Universities individually.

Many professors in Engineering Departments will support financially.

(tuition + living expenses)

GPA and recommendation letters of professors are very important.

Top 10 Korean Universities [Electrical and Computer Engineering]

Top 2	Seoul National University [SNU] (서울대학교) snu.ac.kr Korea Advanced Institute of Science and Technology [KAIST] (한국과학기술원) kaist.ac.kr
Next 8	Seoul National University of Science and Technology [SEOULTECH] (서울과학기술대학교) https://en.seoultech.ac.kr/ Gwangju Institute of Science and Technology [GIST] (광주과학기술원) gist.ac.kr Pohang University of Science and Technology [POSTECH] (포항공과대학교) postech.ac.kr Hanyang University (한양대학교) hanyang.ac.kr Inha University (인하대학교) inha.ac.kr Korea University (고려대학교) korea.ac.kr Sungkyunkwan University (성균관대학교) skku.ac.kr Yonsei University (연세대학교) yonsei.ac.kr

2023 Top 10 USA Universities [Computer Engineering]

https://www.usnews.com/best-graduate-schools/top-engineering-schools?int=top_nav_Engineering

rank	Computer Engineering [Graduate School]
1	Massachusetts Institute of Technology
2	Stanford University
	University of California at Berkeley
4	Carnegie Mellon University
5	Georgia Institute of Technology
	University of Illinois at Urbana Champaign
7	University of Michigan at Ann Arbor
8	Cornell University
	University of Texas at Austin
10	Princeton University
	Purdue University (West Lafayette)
	University of California--San Diego

* Salary in USA

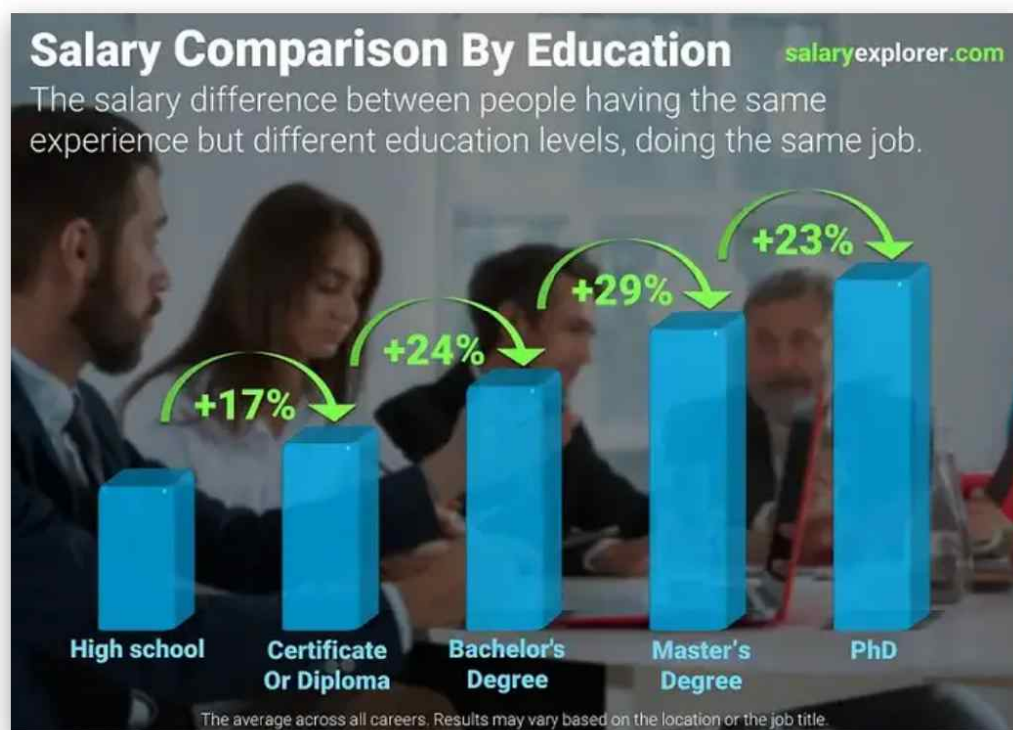
<https://www.mtu.edu/engineering/outreach/welcome/salary/>

* Salary Comparison By Education in Uzbekistan

<https://www.salaryexplorer.com/average-salary-wage-comparison-uzbekistan-engineering-c232f22>

Salary Comparison By Education

How does the education level affect your salary?



100

117

145

187

230